ACTL3182 Wk6

tadhg

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Note

I think it is important to understand some of the finer details, otherwise things dont make sense and mistakes will come !!
I don't know how good these next few slides are (too mathy, not enough mathy) so please lmk!!

Probability revisited

Let Ω be a sample space and $\mathcal{F}$ a collection of events (a σ-algebra).

A probability measure $\mathbb{P}$ needs to satisfy

$$\mathbb{P}(A) \in [0, 1], \quad \forall A \in \mathcal{F},$$
$$\mathbb{P}(\Omega) = 1.$$

Intuitively, the probabilities assigned just need to be between 0 and 1 and the probability of anything happening needs to be 1. Hence, there are multiple possible measures for a single sample space, say $\mathbb{P}$ and $\mathbb{Q}$.

σ-algebra

Recall $\mathcal{F}$ is the collection of events - it is the set of things that we might measure the probability of.

For example, if we roll a D3 and X be the outcome, where $\Omega = \{1, 2, 3\}$ are the possible outcomes. Then, we might have

$$\mathcal{F} = \{\{1\}, \{2\}, \{3\}, \{2, 3\}, \{1, 2\}, \{1, 3\}, \{1, 2, 3\}\}.$$

So, as a probability measure $\mathbb{P}$ is defined for all $A \in \mathcal{F}$, we are able to find out the probability of certain events:

$$\begin{aligned}
\mathbb{P}(X \in \{1\}) &= \mathbb{P}(\text{rolling a 1}), \\
\mathbb{P}(X \in \{1, 3\}) &= \mathbb{P}(\text{rolling odd}), \\
\mathbb{P}(X \in \{1, 2, 3\}) &= \mathbb{P}(\text{rolling anything}).
\end{aligned}$$

$\implies$ $\mathcal{F}$ tells us what questions we can ask about the experiment.

Conditional Expectation

Let X be a random variable. One can think of $\mathbb{E}[X]$ as the "best possible guess" of what X will be.

Similarly, for an event $A \subseteq \Omega$, $\mathbb{E}[X \mid A]$ is our best guess for X given we know A.

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Hence, if $\mathcal{F}$ is a σ-algebra,

$$\mathbb{E}[X \mid \mathcal{F}]$$

is our best guess of X given we can only ask questions from $\mathcal{F}$.

Crucially, $\mathbb{E}[X \mid \mathcal{F}]$ is still random!!!

Motivating Filtrations

The $\mathcal{F}$ we pick dictates how "rich" our probability space is:

- In the previous example, $\mathcal{F}$ contained all possible subsets of Ω, and so $\mathbb{P}$ can tell us anything we want.
- If instead we take

$$\mathcal{G} = \{\{1, 2, 3\}\} \subset \mathcal{F},$$

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Take an expt where the information you have evolves over time (e.g. you observe the price of a stock and keep track of its history). Then, our probability space should get richer as time progresses...

Filtrations

Definition (Filtration)

A filtration $\{\mathcal{F}_t : t \geq 0\}$ is a set of σ-algebras such that

$$s \leq t \implies \mathcal{F}_s \subseteq \mathcal{F}_t.$$

Intuitively, as time progresses the amount of knowledge we have about our process increases.

Martingale

Recall a stochastic process is a series of random variables indexed by t.

Definition (Martingale)

A stochastic process $M(\cdot)$ is said to be a $\mathbb{Q}$-martingale if

$$\mathbb{E}_{\mathbb{Q}}[M(u) \mid \mathcal{F}(t)] = M(t), \quad \forall u \geq t.$$

That is, given all the information up to time t, our best guess of wherever M will be in the future is just what it is right now.

Both sides are random variables!!